ECVOL-bench: An Identity-Controlled Evaluation Protocol and Audit Toolkit for Multimodal Earnings-Call Volatility Prediction

Anonymous ACL submission

Abstract

Since 2019, researchers have predicted post-earnings-call stock volatility from what executives say and how they sound, often reporting gains over past-volatility forecasts. We audit the EC/MAEC label and split files reused across studies. They leave no gap between training and test although targets run 30 days ahead, 44–80% of test calls come from companies seen in training, and day zero sits on the call date although most companies report before the open. We release a protocol and audit toolkit that make these assumptions testable. The protocol recovers company names and dates for an audio corpus that ships without them, takes release times from SEC filings, and provides targets under both horizon definitions and splits disjoint in time and in company. The toolkit runs company-level diagnostics on any prediction file. Under this protocol, open text and audio models show no consistent improvement over past volatility by a margin fixed in advance, and their embeddings mostly identify the company. Five published models rerun under the same checks show the same pattern. Holding companies out raises the text models' error by 12–82% but not the price-only models', and HAR-RV stays ahead of every content model after the cutoff.

1 Introduction

An earnings call is the one regular occasion on which a company's managers speak at length about their business. Since Qin and Yang (2019), a line of papers has used both the transcript and the audio of these calls to predict how volatile the stock will be over the following 3 to 30 trading days, and HTML (Yang et al., 2020), MAEC (Li et al., 2020), VolTAGE (Sawhney et al., 2020a), DialogueGAT (Sang and Bao, 2022), KeFVP (Niu et al., 2023) and ECC Analyzer (Cao et al., 2024b) all report beating classical forecasts. Yu et al. (2025) gave a reason to look again. On a corpus of their own, transcript representations mostly told the model *which company* was speaking, and on that corpus and on the shared benchmarks a forecast from the company's own past volatility matched or beat every transcript model.

What that leaves open is whether any published model keeps its score once the company is taken away, and the files the field scores on cannot answer it. Several studies evaluated on EC and MAEC use redistributed label and split files. We audited them (§3). They leave no gap between training and test dates even though each target runs up to 30 days past its call. Between 44% and 80% of test calls come from companies that are also in training (Figure 1b), and no paper reports what happens when they are not. They define horizons in calendar days, which later papers describe as trading days. They put day 0 on the date of the call, although most companies publish before the open (Figure 1a), a defect noted by Yu et al. (2025) which no shared label set corrects. FinCall-Surprise (Shu et al., 2026a), which ships raw audio, has no ticker, date or speaker fields, so the identity checks cannot be run on it at all. No published model had been rerun under such checks, and no frozen volatility pipeline had been scored on calls after its training cutoff. This paper makes the diagnosis measurable on the files in use and releases the tools for it.[1]

What we contribute. (1) **A protocol over existing corpora**, released as ECVOL-bench. We recover company identities and dates for FinCall-Surprise (92.9% of calls, and a 60-call manual audit passes 60/60), time each release from 8-K filings, and ship targets under both horizon definitions and a forward-looking variant, splits with a 30-session gap, company-disjoint splits and one that applies both rules, and a harness in which every table

[1] Anonymized repository (decision log, harness, checksums, ledger): https://anonymous.4open.science/r/ecvol-bench-E717

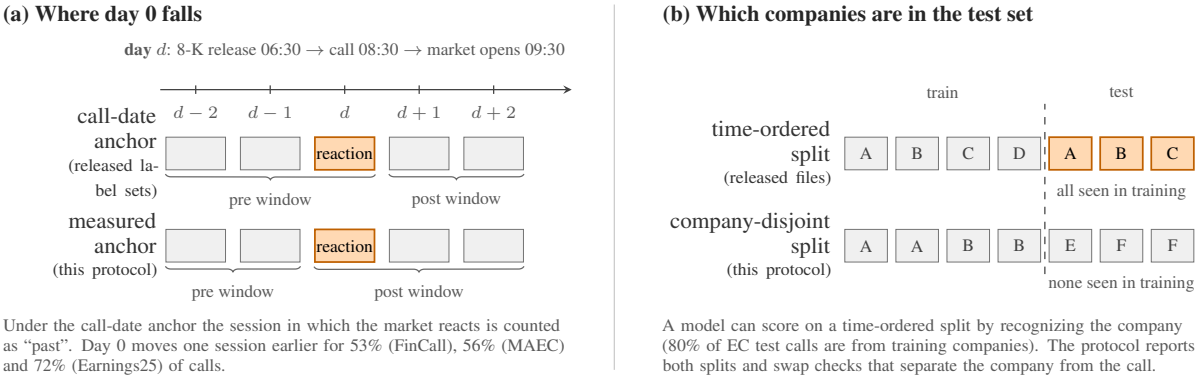


Figure 1: **Two properties of the shared test files.** (a) Most companies release results before the open, so anchoring day 0 on the call date leaves the session in which the market reacts inside the pre window. The protocol anchors on the measured release time. (b) A time-ordered split keeps the same companies on both sides of the boundary. The protocol adds a company-disjoint split and swap checks that make the difference measurable.

comes from one command (§3). It redistributes no raw audio or transcripts. (2) **An audit and a toolkit.** We audit the shared label and split files (Table 1) and the code of seventeen papers, rerun the five runnable models under the same checks, on their own data where it is still available, and release the checks themselves, company-level diagnostics for any prediction file and probes and content swaps for any model one can run. (3) **Results** for modern open-weight text and audio models against a price-only reference, with a margin fixed in advance, and for the frozen pipeline on 475 calls after every model's training cutoff (§5). Stages, checks and tests were fixed in advance in the repository's decision log, whose entries carry git timestamps (history exported in the repository).

2 Related Work

Earnings calls and volatility. Qin and Yang (2019) introduced the task and the EarningsCall corpus, predicting volatility over 3 to 30 days from the transcript and the audio. HTML (Yang et al., 2020) added a sentence-level transformer with a multi-task loss, MAEC (Li et al., 2020) a larger corpus with sentence-aligned audio features, VolTAGE (Sawhney et al., 2020a) and Sawhney et al. (2020b) graph and multimodal fusion, DialogueGAT (Sang and Bao, 2022) a dialogue graph over the Q&A, and KeFVP (Niu et al., 2023) knowledge-enhanced features. ECC Analyzer (Cao et al., 2024b) and Sound of Risk (Chen et al., 2025) are recent positive results and come without code. Most EC and MAEC results in this line are scored on the label and split files that VolTAGE and KeFVP redistribute, which §3 audits. Three newer corpora matter here. FinCall-Surprise (Shu et al., 2026a) ships raw audio and no identities. Earnings25 (Jiang et al., 2026) postdates every model's training data. MM-FinEval (Shu et al., 2026b), which appeared after this study was complete, adds slides and twelve task labels over 2,045 calls with audio, but its release has no ticker or date fields and none of the checks below.

Identity, memorization and lookahead. Yu et al. (2025) showed on a corpus of their own that transcript representations mostly tell the model which company is speaking, and on that corpus, EC and MAEC that past volatility matches or beats every transcript model and that the shipped labels ignore release timing. Language models recover the company even from transcripts with the names removed (Lopez-Lira et al., 2025). Lookahead bias in language models now has diagnostics and cutoff-bounded models of its own (Gao et al., 2025; Yan et al., 2026; Benhenda, 2026), and Gao et al. (2025) include an earnings-call forecasting task. The post-cutoff test in §5 complements that work by scoring frozen volatility pipelines on a later cohort.

Baselines, timing and bias. Persistence, GARCH and HAR-RV are hard to beat (Corsi, 2009; Christensen et al., 2023) and are often missing from the comparisons in this line (Yu et al., 2025, Table T9), so every table here is anchored on them. Yu et al. (2026) anchor the earnings timeline on the press release and time its first tradable moment from news-wire stamps to predict returns. We take the release time from the EDGAR acceptance stamp of the Item 2.02 8-K to place day 0 for volatility, which no label

set in this line had done. Financial-audio studies report demographic imbalance and speech-emotion models carry social biases (Wang et al., 2024; Mathur et al., 2022; Lin et al., 2024), so we report a gender diagnostic.

Reproducibility. Among the score comparisons in the reproduction studies they reviewed, Belz et al. (2021) found about 14% exact matches, and in financial NLP discontinued data APIs and dead repository links make matters worse (Tatarinov et al., 2026).

3 The Shared Test Files and the Protocol

What the field evaluates on. The EarningsCall (EC) benchmark of Qin and Yang (2019) and the two MAEC cohorts are scored on split and label files that VolTAGE and KeFVP redistribute. We audited those files (Table 1). The VolTAGE and KeFVP test files are byte-identical, so the two releases reuse the same label file. Five of the six boundaries between training, validation and test have a zero-day gap (the sixth has three days) while the targets reach 30 days past each call. On the time-ordered splits the field reports, 80.4% (EC), 44.2% (MAEC-15) and 55.7% (MAEC-16) of test calls are from companies that also appear in training. Any time-ordered split allows that, and no paper reports what its model does when the company is held out. The original paper defines its horizons in calendar days, but most later work describes them as trading days (21 sessions rather than 30 at τ=30). Finally, the shipped MAEC labels correlate only 0.52–0.87 at $\tau \geq 7$ (0.25–0.66 at τ=3) with calendar-day targets we recompute from prices under the same call-date anchor, and no single stated rule reproduces them.

Corpora and company identity. FinCall-Surprise (Shu et al., 2026a) is the primary corpus, with 2,688 calls from 2019–2021 and MP3 audio, transcripts and slides, but no ticker, date, CIK or speaker names. We recover the company and the date from the transcript text and the slide metadata, check them against price data, and release the table with its 80 manual corrections. 2,496 calls resolve (92.9%, 95.3% of the earnings-type calls), and a 60-call manual audit passes 60/60. Nine resolved calls (0.4%) carry a pre-2019 date that their transcripts contradict, and all sit in training segments and are flagged in the release. **MAEC** (Li et al., 2020) is the secondary corpus, with 3,443 calls

Benchmark / split	calls	overlap	embargo	label defect
EC (Qin & Yang)	560	80.4%	0 / 0 d	137 zeros
MAEC-15 (KeFVP)	765	44.2%	0 / 0 d	binary cols
MAEC-16 (KeFVP)	1400	55.7%	0 / 3 d	binary cols
ours, FinCall temporal	2200	91.3%	47 / 47 d	n/a
ours, FinCall ticker-disjoint	2433	**0.0%**	n/a	n/a
ours, MAEC temporal	2264	79.4%	101 / 48 d	n/a
ours, MAEC ticker-disjoint	2688	**0.0%**	n/a	n/a

Table 1: **Audit of the shared test files** (regenerated by `ecvol audit substrate` from checksummed copies of the VolTAGE and KeFVP files, whose test files are byte-identical). Overlap is the share of test calls whose company is in training, and embargo is the gap train→val / val→test. Label defects: EC's single-day volatility series holds 137 exact zeros, and the MAEC files interleave binary 0/1 columns among the volatility labels. Every time-ordered split allows overlap, ours included, but no paper in this line reports the disjoint case.

from 2015–2018 and sentence-aligned audio features. The release we could access ships no raw audio. **Earnings25** (Jiang et al., 2026) has 514 S&P 500 calls in the release we ingested, 501 of them from October to December 2025, with webcast MP3 at 64 kbps for 281 calls and at most 24 kbps for 232. 476 of its calls map to a ticker and join to prices, and 475 carry a target.

Targets. Daily adjusted prices come from Yahoo Finance through the yfinance package, whose terms do not permit redistribution, and were checked against Tiingo on 50 tickers. We release derived volatility targets, never the price series. Let r_t be the simple return of the adjusted close on session t, counted from day 0, the last session before the call's information is public. The main target is $v_{\text{post}}(\tau) = \ln \sqrt{\frac{1}{n}\sum_{t=1}^{n}(r_t - \bar{r})^2}$ for $\tau \in \{3, 7, 15, 30\}$, and $v_{\text{pre}}(\tau)$ is the same quantity over the sessions ending at day 0. Under the **trading-day** definition $n = \tau$. Under the **calendar-day** definition of Qin and Yang (2019) the window holds every session within τ calendar days, so $n \approx 0.7\tau$. We drop any window with fewer than two sessions and record the reason, which removes 42.5% of FinCall's calls at τ=3 calendar days. v_{post} correlates only 0.78 between the definitions at τ=3 and the persistence MSE is 44–63% higher under calendar days at τ=3 (31–41% at longer horizons), so a published MSE means little unless its definition is stated (Appendix Table T7). Every result below uses trading days, and the definition is a column in every result table. We add two targets that are harder to game with company identity. The **change** $\Delta v = v_{\text{post}} - v_{\text{pre}}$ subtracts the

company's own level, and the **HAR-residual** subtracts a HAR-RV forecast. After joining to prices, FinCall has 2,433 calls with a target (97.5%) and MAEC 2,688 of 3,443 (78.1%, the rest lacking usable price history).

When the information became public. Yu et al. (2025) showed that most calls in EC and MAEC are released before the open and that the shipped labels ignore this, which their DEC corpus corrects with a vendor flag. Almost no transcript states a clock time, so we use the EDGAR acceptance time of the earliest 8-K filing with Item 2.02 (the earnings release) within three days before to one day after the call as a proxy for the release time. That covers 89.3% of FinCall, 68.6% of MAEC and 99.8% of Earnings25. The rest fall back, in recorded order, to Earnings25's call-start time, the before/after-market flag of Yu et al. (2025), and an after-hours assumption. A release before the open makes the call date itself day 1. Compared with the call-date anchor that every released label set uses, **day 0 moves for 53% of FinCall, 56% of MAEC and 72% of Earnings25 calls**, almost always one session earlier. Under the published anchor, the session in which the market reacts therefore sits in the *pre* window for most calls, and Δv gains a positive mean on the measured anchor ($+0.51$ at $\tau{=}3$ on FinCall). The call-date targets ship as a labeled variant. On Earnings25, which records call start times, the filing precedes the call for 98.9% of calls (median 100 minutes). A press release can precede its 8-K. Filings accepted within an hour after the open, where an earlier release could move day 0, are 1.1% of FinCall, 3.1% of MAEC and 0.8% of Earnings25 calls, a sensitivity diagnostic rather than an upper bound on the disagreement. The call can still be running during session 1, so these targets describe a release-aligned event window. A forward-looking variant starts the post window one session later, so that no target return begins before the call has ended (true for all 476 Earnings25 calls with a recorded start time).

Splits and release. *Temporal* splits (about 70/10/20 in date order) leave a 30-trading-day gap between segments so that no target window crosses a boundary. *Ticker-disjoint* splits keep every validation and test company out of training. Their overlap is 0.0%, against 91.3% (FinCall) and 79.4% (MAEC) on our own temporal splits. Because they are not in date order, a *date-ordered company-disjoint* split applies both rules at once, at the cost of small test and validation sets (97 and 22 FinCall calls). Appendix Table T5 gives calls, companies, overlaps and dates for every split. The harness fails any feature that reads price data after the call's as-of time, and leakage assertions run in CI. The release contains all of the above, the derived-feature archives, the harness, and every result CSV with its run record (resolved config, git commit, seeds and file hash).

4 Models and Checks

Models, from simplest to richest. We organize our own models in stages. **Stage 0** holds the classical forecasts, namely persistence ($\hat{v} = v_{\text{pre}}$), the training-set mean of the target (a constant), EWMA with $\lambda{=}0.94$ (J.P. Morgan and Reuters, 1996), HAR-RV (Corsi, 2009) and GARCH(1,1) (Bollerslev, 1986). **Stage 1** is a LightGBM model (Ke et al., 2017) on past-volatility features, transcript length, number of Q&A turns and a **company fixed effect**, a price-and-metadata baseline with ticker features. **Stage 2** uses the transcript through BGE-M3 (Chen et al., 2024) embeddings pooled per section (1024-d), FinBERT (Araci, 2019) sentiment aggregates and surface statistics. **Stage 3** uses the audio, resampled to 16 kHz mono. Its features are eGeMAPSv02 functionals (Eyben et al., 2016) extracted with openSMILE 2.6.0 (Eyben et al., 2010), and WavLM-Large (Chen et al., 2022) and emotion2vec+ (Ma et al., 2024) embeddings over 30-second windows, averaged per call. **Stage 4** combines modalities by gated fusion or by stacking (a ridge over the text, audio and Stage-1 predictions, fit on the validation set). From Stage 2 on, every model is reported with and without a block of past-volatility features (v_{pre} at all horizons plus the HAR-RV inputs). The prediction heads are a ridge regression, with its penalty chosen on validation, and a small MLP run with five seeds. The MLPs are unstable across seeds (standard deviation of R^2_{OOS} up to 6.8), so the main grid uses one deterministic model per stage, chosen by a rule recorded in the decision log. MLP cells are in Appendix A.

Metrics and tests. We report MSE on log realized volatility and, as the headline metric, R^2_{OOS} relative to persistence, following Gu et al. (2020) with persistence in place of their zero forecast, defined as $1 - \sum(y - \hat{y})^2 / \sum(y - \hat{y}_{\text{persist}})^2$. Because Δv has a positive mean on the measured anchor, persistence (which predicts $\Delta v{=}0$) is a weak reference there, so every Δv claim is also com-

pared with the constant model. Diebold–Mariano tests (Diebold and Mariano, 1995) with the small-sample correction of Harvey et al. (1997) compare per-call squared errors against persistence and against the previous stage. The confirmatory comparisons (each stage against Stage 1 on Δv across horizons, Stage 4 against the best single modality, and the published split against the controlled split for each reproduced model) are Holm-corrected (Holm, 1979). The per-head asterisks in Tables T2 and T3 are uncorrected. The central question is whether call content improves on the same model without it, so each content model is also compared, on the same calls, with the past-volatility ridge. For Stages 2 and 3 that ridge is the same model without content. For the Stage-4 stack it is a common price-only reference, not an exact ablation. We report the relative MSE reduction with a 95% bootstrap interval that resamples companies and read it against a 2% margin fixed before the comparison was run, a practical threshold below most gaps between the classical baselines (Appendix Table T1). An interval above the margin is an improvement, one below it shows no meaningful benefit, and anything else is inconclusive. All other analyses are exploratory.

Identity checks. We use four. A *company-only* model predicts each company's training mean and shows how far identity alone gets. Two *swaps* replace each test call's transcript or audio, either with another call by the same company or with a random call from the corpus. If the same-company swap barely moves a model's score, the model may be reading what is stable about the company rather than the call. If the random swap barely moves it either, the swapped modality contributes little that this test can detect. A linear *probe* tries to recover the ticker from the frozen call embedding. Finally, every table shows company-disjoint results next to temporal ones. The command `ecvol audit predictions` runs the prediction-side versions of the company-only and swap checks on any file with one row per (call, ticker, date, split, horizon, truth, prediction), and reports split integrity, seen versus unseen companies, the between-company share of prediction variance and company-clustered significance. That is a descriptive diagnostic (Appendix Table T6). A permutation of existing predictions moves the price inputs along with the content, so it is not a substitute for the input-level swaps, which need the model itself.

Reproductions and post-cutoff tests. Of seventeen papers in this line, one releases code that runs as released (Same-Company-Same-Signal), seven release code that needs repairs or inputs that were never shipped, and nine release nothing usable (Table T13). We rerun the five with usable code (HTML, KeFVP, Same-Company-Same-Signal, Sawhney et al., 2020b, DialogueGAT) *on their own data*, using the authors' code wherever it can be made to start and a line-by-line port where it cannot, with every patch in a released ledger. We compare with the published number and then rerun the same model under a gapped and a ticker-disjoint split of the same data. Earnings25 allows two more tests, the frozen Stage-3 audio models rerun by bitrate, and the whole FinCall-trained pipeline scored, with no retraining, on calls that postdate every model's training cutoff.

5 Results

The tables in this section use FinCall-Surprise and the trading-day definition, on the Δv target except Table 5, which uses the level. MAEC, the level target, MSE, the calendar-day definition and every other model are in Appendix A and in the supplement, regenerated from the released CSVs.

The baseline to beat. HAR-RV is strong everywhere except at τ=30 on the two date-ordered splits (Table 2), whose training periods include the COVID spike while the test year (2021) is calm. There, every long-horizon forecast of the volatility level does worse than persistence, ours and every content model alike. The same models do fine on the ticker-disjoint split and on MAEC, so we treat this as a regime effect and note it. GARCH(1,1) beats HAR-RV at every temporal horizon, and HAR-RV beats it on the ticker-disjoint split (Appendix Table T1). The reference for every content claim below is the past-volatility ridge S0′, which sees the same price inputs as each content model and nothing else. GARCH is not that reference, since it is a different estimator on different inputs.

Text models mostly read the company (Table 3, Figure 2). Adding text features to the past-volatility ridge makes it *worse* in all eight cells of Table 2. The paired test finds no meaningful benefit in six and no improvement in any. A linear probe recovers the ticker from the frozen call embedding 89.5% of the time on FinCall and 53.9% on MAEC. On the temporal split, where test com-

Model	τ=3	τ=7	τ=15	τ=30
Temporal split (n=483)				
S0 train mean	0.165	0.158	0.060	0.088
S0 HAR-RV	0.444	0.372	0.138	−0.285
S0′ past-vol ridge	0.412	0.314	0.094	0.113
S1 GBDT (ticker FE)	0.340	0.249	−0.126	−0.948
S2 text ridge + vol	0.333†	0.285	0.080	−0.077†
S3 audio ridge + vol	0.353†	0.268†	0.033†	0.050†
S4 stack fusion	0.406	0.347	0.066	0.100
Date-ordered disjoint split (n=97)				
S0 train mean	0.152	0.147	0.080	0.119
S0 HAR-RV	0.446	0.384	0.209	−0.120
S0′ past-vol ridge	0.409	0.315	0.159	0.158
S1 GBDT (ticker FE)	0.300	0.047	−0.324	−1.063
S2 text ridge + vol	0.160†	0.169†	0.039†	−0.056†
S3 audio ridge + vol	0.312†	0.147†	0.057†	0.098
S4 stack fusion	0.339†	0.108†	−0.180†	−0.028†

Table 2: **The ladder on FinCall-Surprise.** R^2_{OOS} vs. persistence on Δv (test). Persistence is 0 by construction. The second block keeps test companies out of training and the split in date order. One canonical model per stage (the Stage-3 ridge uses WavLM and eGeMAPS). Each content cell is tested against the past-volatility ridge S0′ on the same calls. A dagger marks a 95% interval of the relative MSE reduction, resampled by company, that lies below the 2% margin fixed in advance, an asterisk one that lies above it, and an unmarked cell is inconclusive (Appendix Table T4). The full grid, MAEC, level-v, MSE and the MLP heads are in Appendix A.

Identity probe (linear, frozen call embedding)	accuracy	×chance
BGE-M3 text, FinCall (357 tickers)	89.5%	319×
BGE-M3 text, MAEC (902 tickers)	53.9%	487×
WavLM audio, FinCall (356 tickers)	76.3%	271×
emotion2vec+ audio, FinCall (356 tickers)	29.4%	105×

Content swap (R^2_{OOS} on Δv)	τ=3	τ=7	τ=15	τ=30
Text ridge, temporal split				
real	0.182	0.202	0.044	−0.133
same-company swap	0.174	0.194	0.047	−0.110
random swap	0.145	0.110	0.008	−0.152
Text ridge, date-ordered company-disjoint				
real	0.122	0.133	0.009	−0.092
same-company swap	0.136	0.139	0.019	−0.057
random swap	0.173	0.166	0.084	0.099

Table 3: **Identity checks.** Frozen embeddings identify the company far above chance. Where companies recur (temporal), giving a test call another call by the *same* company barely changes the text model, while a random call hurts. On the date-ordered company-disjoint split the real transcript scores no better than either swap. The audio swaps are in Appendix Table T3.

panies also appear in training, giving each test call the transcript of *another call by the same company* changes the score by less than 0.03, while a random transcript hurts. On the ticker-disjoint split the same-company swap does hurt at $\tau \geq 15$ (supplement), a sign of some call-specific sensitivity, but that split is not in date order. On the date-ordered company-disjoint split the real transcript scores below both swaps, and the random swap lands near the constant (0.17 against 0.15 at τ=3), so real text from an unseen company moves predictions the wrong way. On MAEC the same-company swap also moves the score by 0.02 or less, and adding text to the past-volatility ridge lowers R^2_{OOS} in all twelve cells of its three splits.

Audio shows no consistent gain over past volatility. Audio-only ridges on eGeMAPS, WavLM-Large and emotion2vec+ stay within 0.10 of the constant at every horizon (Appendix Table T3) and never beat it by more than 0.02. Adding WavLM and eGeMAPS to the past-volatility ridge makes it worse in all eight cells, with no meaningful benefit in seven. In the audio-swap check (Appendix Table T3), the model with past volatility keeps its score at τ=3 when every test call is given a *random other company's* audio (0.456 to 0.453). At longer horizons the swap costs more, but the model with real audio stays below the ridge with past volatility alone. A probe still recovers the ticker from pooled WavLM embeddings 76.3% of the time. A pitch-proxy gender diagnostic finds a small gap (test MSE 0.243 against 0.239). Rerun on Earnings25 by bitrate, the gain from real over randomly swapped audio never exceeds 0.03, is negative in 7 of 12 cells, and is no larger at 64 kbps than at lower bitrates (Appendix Table T11), a consistency across observed groups rather than a controlled codec test.

Combining modalities. Gated fusion is below the ridge heads with past volatility in 15 of 16 temporal and ticker-disjoint cells. The stacking model is the best content model. Against the past-volatility ridge it shows no meaningful benefit in all four date-ordered company-disjoint cells and is inconclusive in the four temporal ones. Its one apparent gain is on the ticker-disjoint split at τ=30, which is not in date order. There it reduces MSE by 11% against the past-volatility ridge, with an interval from 1% to 20% that is inconclusive under the margin, and by 8% against HAR-RV, with an interval that includes zero (Appendix Table T4). The forward-looking target lifts that cell above the margin, while on the date-ordered split the same model is 25% worse. Our criterion for a consistent multimodal improvement, fixed in advance as gains surviving the checks on at least two of four

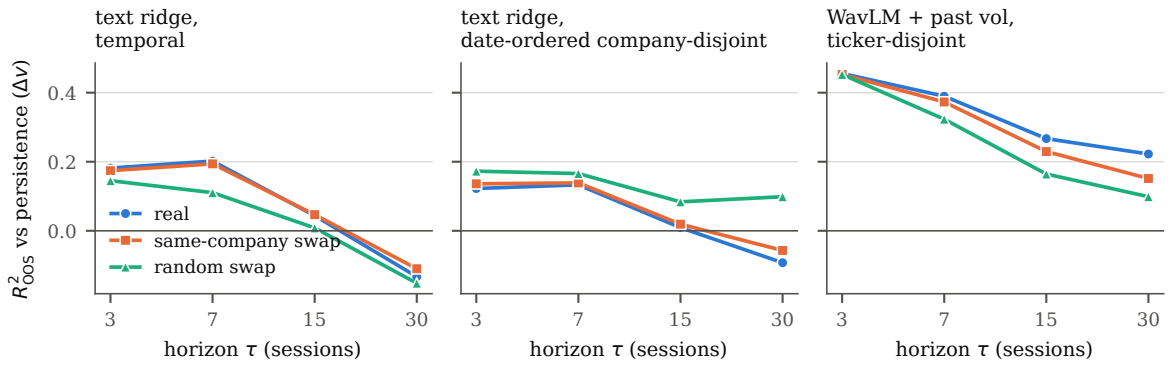


Figure 2: **Content swaps.** R^2_{OOS} vs. persistence on Δv (test) when each test call keeps its own transcript or audio (real), receives another call by the same company, or receives a random call. Where companies recur in training (left), the same-company swap leaves the text model unchanged. On the date-ordered company-disjoint split (middle) the real transcript does no better than either swap. The audio model with past volatility (right, ticker-disjoint split) keeps its score under a random swap at τ=3. Data from Table 3 and Appendix Table T3.

horizons, was not met.

Published models under the checks (Table 4, Figure 3). Among the five papers we could run, the results that reproduce belong to price-only or same-company models (Appendix A.6 separates direct reproductions, repaired runs and ports). Same-Company-Same-Signal reproduces from its shipped data, and KeFVP, a price-series model conditioned on text, reproduces seven of eight MAEC cells. HTML's released code does not reproduce its published table, and the published ensemble of Sawhney et al. (2020b) is matched only through its finance branch, with mixing weights chosen on the test set. DialogueGAT's corpus was never released, so it is ported to FinCall. Holding companies out raises the MSE of the three text models (HTML, the ensemble and the TMLP text model of Same-Company-Same-Signal, SCSS) by 12–82%, while SCSS's price-only TSMixer and KeFVP move by -10% to $+7\%$ with no consistent sign. The SCSS and KeFVP comparisons use the same test calls. The HTML and ensemble ones do not, so they also reflect a change of test set. Training-set size is an unlikely explanation (Table 4), though only a matched removal of unrelated companies would settle it. The training-test gap moves models by at most 13%. HTML's own trained model changes by at most 4% when a test transcript is swapped for another call by the same company, against 9–23% for a different company's call.

After the cutoff (Table 5). We scored the FinCall-trained models, unchanged, on 475 Earnings25 calls from late 2025, after every model's training cutoff. 280 of those companies (59%) appear in FinCall's training split four years earlier, so this is a later cohort, not an unseen one. On the level target, no model that was competitive before the cutoff got worse. HAR-RV rose from 0.44, 0.37, 0.14 and -0.29 to 0.60, 0.54, 0.35 and 0.08, and the content ridges rose with it (only the fusion ridge is unstable, at τ=30). Levels are not comparable across the cutoff. HAR-RV remains above every content model at every horizon on both sides of it. Refit under the call-date anchor of the published label sets, the same models collapse before the cutoff. Text plus past volatility on Δv falls to

Model (data), change in test MSE	τ=3	τ=7	τ=15	τ=30
Companies held out (ticker-disjoint vs. the model's own split)				
HTML text (EC)	+18%	+55%	+55%	+82%
HTML text+audio (EC)	+20%	+44%	+61%	+70%
Sawhney ensemble (EC)	+13%	+28%	+41%	+38%
SCSS TMLP, text (DEC)	+12%	+23%	+25%	+27%
SCSS TSMixer, price (DEC)	+7%	+5%	−6%	−8%
KeFVP, text+price (MAEC-15)	−3%	+4%	+3%	−1%
KeFVP, text+price (MAEC-16)	−9%	−3%	−10%	−4%
30-session embargo vs. the model's own split				
HTML text (EC)	+10%	+3%	−3%	+11%
Sawhney ensemble (EC)	+5%	−6%	−2%	−9%
SCSS TMLP (DEC)	−4%	+1%	0%	+3%
HTML text, transcript replaced by (own split, 97 calls)				
another call, *same* company	+1%	+4%	+3%	+2%
a call of a *different* company	+9%	+11%	+21%	+23%

Table 4: **The field's runnable models under our controls.** Each row keeps the configuration of Appendix Table T9 and changes only the split. Holding companies out costs every text model 12–82% while the price-dominated models show no consistent change, with held-out training sets as large (EC) or cut by the same third (DEC, MAEC). Absolute values are in Appendix Table T10.

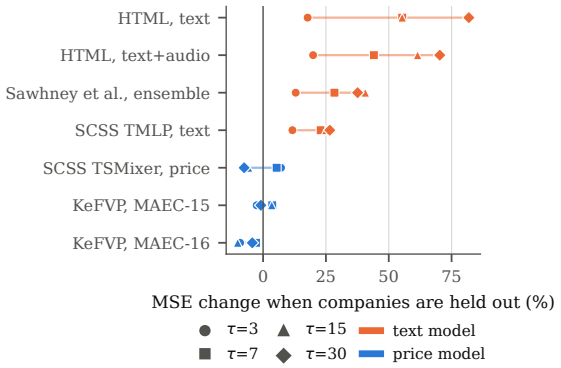


Figure 3: **Published models with companies held out.** Change in test MSE when the same model is scored on a ticker-disjoint split of its own data, one marker per horizon (data from Table 4). The MSE of the three text models rises by 12–82%, and the two price-based models show no consistent change.

Model	cutoff	τ=3	τ=7	τ=15	τ=30
S0 train mean	before	0.36	0.14	−0.37	−1.05
	after	0.58	0.47	0.15	−0.29
S0 HAR-RV	before	0.44	0.37	0.14	−0.29
	after	0.60	0.54	0.35	0.08
S2 text ridge + vol	before	0.43	0.34	0.03	−0.45
	after	0.57	0.50	0.25	−0.20
S3 WavLM ridge + vol	before	0.38	0.24	−0.09	−0.66
	after	0.57	0.43	0.15	−0.32
S4 fusion ridge + vol	before	0.41	0.32	−0.02	−0.63
	after	0.50	0.46	0.20	−1.93

Table 5: **Frozen-pipeline post-cutoff evaluation.** Heads fit once on FinCall's training split and scored unchanged before the cutoff (the 2021 FinCall test year) and after it (475 Earnings25 calls, late 2025), on the level target, R^2_{OOS} vs. persistence. HAR-RV stays above every content model at every horizon on both sides of the cutoff. Bootstrap confidence intervals and the Δv rows are in the released CSV.

between −1.7 and −3.6 at 3 to 15 days (Appendix Table T12), because the reaction session then sits inside the past-volatility features.

6 Discussion

Across every model we trained, the same picture holds. Text features do not improve on past volatility alone, and the swap checks say why. A probe reads the ticker out of the call embedding nine times in ten, and where a company recurs in training, handing the model a different call by that company moves its score by less than 0.03. Audio identifies the company without improving the forecast, and the one stacking gain over the price-only reference appears only on a split that is not in date order. The shared test files could not have shown any of this, because they carry no company identity for the audio corpus, no company-disjoint split and no measured day 0. The controlled reruns show large company sensitivity for HTML, the Sawhney ensemble and the SCSS text model, no consistent change for the price-dominated models, and an inconclusive port for DialogueGAT. Two recent positive results (Chen et al., 2025; Cao et al., 2024b) release no code and could not be checked. Transcripts and audio may well carry information about a company's risk. The test files in use cannot separate that information from knowing which company is speaking. Once they can, under the temporal and date-ordered company-disjoint protocols none of the content models evaluated here improves on its past-volatility reference by the 2% margin fixed in advance, and the isolated gains that remain need targeted validation. That is the bar the next model has to clear.

7 Recommendations and Outlook

The results above say what an evaluation has to control for before a content effect can be claimed, not that earnings calls carry no information. We suggest that future work on this task report six things. A **company-disjoint** split next to the temporal one. A **gap** of at least the longest horizon between training and test. **Past-volatility baselines** as the reference for every claim. The **same-company and random swaps** when the model is available, otherwise the prediction-only diagnostics of the released audit command, reported as descriptive. **Day 0 measured** from the release time. And the **horizon definition** with every number. Appendix Table T14 gives the reason for each and where the release provides it. ECVOL-bench covers all six for FinCall-Surprise and MAEC, and a quickstart script produces the baseline, a prediction file and the audit report from a released archive.

Content signal may still be found once it is looked for in the right place. Measured release times make **event-window targets** possible, where language about guidance and surprise should matter most and same-company history explains least. **Structured features** such as the transcript dimensions of Yu et al. (2026) can be audited one field at a time and tested for what they add beyond the numbers. **Audio** deserves a test on source-quality recordings with resolved speakers, and **additional corpora** such as MM-FinEval (Shu et al., 2026b) are open to the identity recovery used here.

Limitations

One post-cutoff period. The post-cutoff test rests on 475 calls from five calm months of 2025, 470 of them from October to December. Period, companies and source all change at once, so it is a temporal external validation of the ordering of models, not a measurement of pretraining contamination, and 59% of its companies appear in FinCall's training split. **Small date-ordered sets.** The split that is both date-ordered and company-disjoint keeps 97 FinCall test calls and only 22 FinCall and 5 MAEC validation calls, on which ridge penalties are chosen and the stack is fit, so model selection and stacking are least stable there. The intervals resample companies but not calendar time, so shocks common to one quarter are not accounted for. **Forward-looking target.** It assumes a call after a pre-open release ends before that day's close, checked on Earnings25 only, and it was run on FinCall for the canonical models only. **Reproductions run on their own data**, not on ECVOL-bench, to keep "the model fails" separable from "the model does not transfer". Two carry known costs. KeFVP's released EC embeddings are unavailable upstream, and DialogueGAT's corpus was never released, so our port stops at its protocol. **MLP heads are seed-unstable** (standard deviation of R^2_{OOS} across seeds up to 6.8 on the level target and 1.1 on Δv), which is why the grid uses deterministic models. MLP cells in the appendix are seed means with that caveat. **Short-horizon R^2_{OOS} inflation.** The persistence denominator is noisiest at τ=3 (its MSE is about 2.8 times that at τ=7), so short-horizon gains are easier to show. **Timestamps.** 10.7% of FinCall and 31.4% of MAEC calls (13.0% of the MAEC calls that carry targets) have no Item 2.02 filing in the window and use a recorded fallback tier, and intraday releases are treated as after-hours. Results under the call-date anchor of prior label sets are not interchangeable with ours, and the call-date targets are shipped as a variant. **Regime sensitivity.** FinCall-Surprise spans COVID. The temporal τ=30 failure shows that long-horizon level forecasts are regime-sensitive, which is why we emphasize Δv and per-year breakdowns. **MAEC coverage.** Its 78.1% price-join rate (2015–2018 delistings) makes it a secondary corpus, and it ships no raw audio. **Source audio.** Every FinCall file is a 40 kbps MP3 transcode and Earnings25 is 64 kbps or lower, so no result here is established on lossless audio. **No diarization.** Audio is pooled per call, and the gender analysis uses a dominant-speaker pitch proxy, so a small gap on that proxy does not establish the absence of one. **Identity reconstruction is imperfect.** 7.1% of FinCall calls remain unresolved and are excluded with a reason code, and nine resolved training calls carry a date their transcript contradicts. **No fine-tuned or LLM-based content model of our own.** Stages 2 to 4 use frozen encoders with ridge and MLP heads. The HTML reproduction, a trained hierarchical transformer over frozen BERT features, shows the same dependence on company identity (Table 4), so the pattern is not an artifact of a linear head. Whether a fine-tuned encoder behaves differently is untested here. §7 points to structured features extracted by language models as the place to look. The audit command lets anyone check such a model from its prediction file.

Ethical Considerations

Data and licensing. We use openly licensed corpora and open-weight models. FinCall-Surprise is released under Apache-2.0, which its authors state covers the transcripts, audio and slides (Shu et al., 2026a, Appendix A), and which they describe as collected from corporate websites and public reference platforms. We rely on that license for research use, redistribute none of the raw audio or transcripts, and release only identity and timing tables, targets, splits and pooled per-call features, each carrying the Apache-2.0 notice and attribution. The transcripts name the companies in the text, so the identity table records public facts about public companies rather than restoring a withheld field, and we release no speaker-level audio or names. MAEC is CC-BY-SA-4.0 and its derived archives are released under the same license. Earnings25 releases its transcripts and metadata under CC-BY-4.0. Its audio is redistributed subject to the original providers' terms, and we release only pooled per-call features from it. Raw audio or transcripts from sources with unclear terms (EC, DEC, the reproduced repositories) are never redistributed. Release times come from EDGAR, which is public-domain US government data. Daily prices come from Yahoo Finance through the yfinance package, whose terms do not permit redistribution, so we release derived volatility targets and never the price series. **Gender-related bias.** Prior financial-audio studies have reported gender-related prediction biases (Wang et al., 2024; Mathur et al., 2022), and

speech-emotion models carry social biases (Lin et al., 2024). Motivated by these findings, we report an exploratory comparison of prediction errors using a dominant-speaker pitch proxy. This diagnostic does not establish the gender composition of our corpora, and it is not a full assessment of gender-related fairness. **Intended use.** This is a research tool, not investment advice. Predictions are noisy and regime-sensitive and should not drive automated trading. **Reproducibility.** Given the documented reproducibility problems in financial NLP (Tatarinov et al., 2026), the environment is pinned, every data file is checksummed, every result CSV regenerates byte for byte from its command and run manifest, and the curated tables copy cells from those CSVs.

References

Dogu Araci. 2019. FinBERT: Financial sentiment analysis with pre-trained language models. arXiv:1908.10063. Model: huggingface.co/ProsusAI/finbert.

Anya Belz, Shubham Agarwal, Anastasia Shimorina, and Ehud Reiter. 2021. A systematic review of reproducibility research in natural language processing. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume*, pages 381–393. Association for Computational Linguistics.

Mostapha Benhenda. 2026. Look-ahead-bench: A standardized benchmark of look-ahead bias in point-in-time LLMs for finance. arXiv:2601.13770.

Tim Bollerslev. 1986. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327.

Yupeng Cao, Zhi Chen, Prashant Kumar, Qingyun Pei, Yangyang Yu, Haohang Li, Fabrizio Dimino, Lorenzo Ausiello, K. P. Subbalakshmi, and Papa Momar Ndiaye. 2024a. RiskLabs: Predicting financial risk using large language model based on multimodal and multi-sources data. arXiv:2404.07452. Workshop on Multimodal Financial Foundation Models (MFFM) at ICAIF 2024.

Yupeng Cao, Zhi Chen, Qingyun Pei, Nathan Lee, K. P. Subbalakshmi, and Papa Momar Ndiaye. 2024b. ECC Analyzer: Extracting trading signal from earnings conference calls using large language model for stock volatility prediction. In *Proceedings of the 5th ACM International Conference on AI in Finance*, ICAIF '24, page 257–265, New York, NY, USA. Association for Computing Machinery.

Jianlyu Chen, Shitao Xiao, Peitian Zhang, Kun Luo, Defu Lian, and Zheng Liu. 2024. M3-embedding: Multi-linguality, multi-functionality, multi-granularity text embeddings through self-knowledge distillation. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 2318–2335, Bangkok, Thailand. Association for Computational Linguistics.

Sanyuan Chen, Chengyi Wang, Zhengyang Chen, Yu Wu, Shujie Liu, Zhuo Chen, Jinyu Li, Naoyuki Kanda, Takuya Yoshioka, Xiong Xiao, Jian Wu, Long Zhou, Shuo Ren, Yanmin Qian, Yao Qian, Jian Wu, Michael Zeng, Xiangzhan Yu, and Furu Wei. 2022. WavLM: Large-scale self-supervised pre-training for full stack speech processing. *IEEE Journal of Selected Topics in Signal Processing*, 16(6):1505–1518. Model: huggingface.co/microsoft/wavlm-large.

Xiaoliang Chen, Xin Yu, Le Chang, Teng Jing, Jiashuai He, Ze Wang, Yangjun Luo, Xingyu Chen, Jiayue Liang, Yuchen Wang, and Jiaying Xie. 2025. The sound of risk: A multimodal physics-informed acoustic model for forecasting market volatility and enhancing market interpretability. arXiv:2508.18653.

Kim Christensen, Mathias Siggaard, and Bezirgen Veliyev. 2023. A machine learning approach to volatility forecasting. *Journal of Financial Econometrics*, 21(5):1680–1727.

Fulvio Corsi. 2009. A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2):174–196.

Francis X. Diebold and Roberto S. Mariano. 1995. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263.

Florian Eyben, Klaus R. Scherer, Björn W. Schuller, Johan Sundberg, Elisabeth André, Carlos Busso, Laurence Y. Devillers, Julien Epps, Petri Laukka, Shrikanth S. Narayanan, and Khiet P. Truong. 2016. The Geneva minimalistic acoustic parameter set (GeMAPS) for voice research and affective computing. *IEEE Transactions on Affective Computing*, 7(2):190–202.

Florian Eyben, Martin Wöllmer, and Björn Schuller. 2010. openSMILE – the Munich versatile and fast open-source audio feature extractor. In *Proceedings of the 18th ACM International Conference on Multimedia*, pages 1459–1462. Association for Computing Machinery.

Zhenyu Gao, Wenxi Jiang, and Yutong Yan. 2025. Detecting lookahead bias in LLM forecasts. arXiv:2512.23847.

Shihao Gu, Bryan Kelly, and Dacheng Xiu. 2020. Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5):2223–2273.

David Harvey, Stephen Leybourne, and Paul Newbold. 1997. Testing the equality of prediction mean squared errors. *International Journal of Forecasting*, 13(2):281–291.

Sture Holm. 1979. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70.

Denglin Jiang, Haoran Zhou, Anshul Wadhawan, Brendan Fahy, Vinay Ramesh, David Weisberg, Dmitriy Derkachevskiy, Helen Sheehan, Srivas Prasad, and Michele Franceschini. 2026. Earnings25: A comprehensive 500-hour speech benchmark for finance. arXiv:2607.23813. Data: Zenodo record 10.5281/zenodo.18762168.

J.P. Morgan and Reuters. 1996. RiskMetrics—technical document. Technical report, J.P. Morgan/Reuters, New York.

Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. 2017. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems 30*, pages 3146–3154.

Jiazheng Li, Linyi Yang, Barry Smyth, and Ruihai Dong. 2020. MAEC: A multimodal aligned earnings conference call dataset for financial risk prediction. In *Proceedings of the 29th ACM International Conference on Information and Knowledge Management (CIKM)*, pages 3063–3070.

Yi-Cheng Lin, Haibin Wu, Huang-Cheng Chou, Chi-Chun Lee, and Hung-yi Lee. 2024. Emo-bias: A large scale evaluation of social bias on speech emotion recognition. In *Proceedings of Interspeech 2024*, pages 4633–4637.

Mengpu Liu, Mengying Zhu, Xiuyuan Wang, Guofang Ma, Jianwei Yin, and Xiaolin Zheng. 2024. ECHO-GL: Earnings calls-driven heterogeneous graph learning for stock movement prediction. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 13972–13980.

Yingnan Liu, Ningbo Bu, Zhiqiang Li, Yongmin Zhang, and Zhenyu Zhao. 2025. AT-FinGPT: Financial risk prediction via an audio-text large language model. *Finance Research Letters*, 77:106967.

Alejandro Lopez-Lira, Yuehua Tang, and Mingyin Zhu. 2025. The memorization problem: Can we trust LLMs' economic forecasts? arXiv:2504.14765.

Ziyang Ma, Zhisheng Zheng, Jiaxin Ye, Jinchao Li, Zhifu Gao, Shiliang Zhang, and Xie Chen. 2024. emotion2vec: Self-supervised pre-training for speech emotion representation. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 15747–15760. Association for Computational Linguistics. Model: huggingface.co/emotion2vec/emotion2vec_plus_large.

Puneet Mathur, Mihir Goyal, Ramit Sawhney, Ritik Mathur, Jochen Leidner, Franck Dernoncourt, and Dinesh Manocha. 2022. DocFin: Multimodal financial prediction and bias mitigation using semi-structured documents. In *Findings of the Association for Computational Linguistics: EMNLP 2022*, pages 1933–1940, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.

Hao Niu, Yun Xiong, Xiaosu Wang, Biao Yang, and Yao Zhang. 2024. How does textual information selection influence time series forecasting? A cross-modal perspective on financial volatility prediction. In *2024 IEEE International Conference on Multimedia and Expo (ICME)*, pages 1–6. IEEE.

Hao Niu, Yun Xiong, Xiaosu Wang, Wenjing Yu, Yao Zhang, and Weizu Yang. 2023. KeFVP: Knowledge-enhanced financial volatility prediction. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 11499–11513, Singapore. Association for Computational Linguistics.

Yu Qin and Yi Yang. 2019. What you say and how you say it matters: Predicting stock volatility using verbal and vocal cues. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 390–401, Florence, Italy. Association for Computational Linguistics.

Yunxin Sang and Yang Bao. 2022. DialogueGAT: A graph attention network for financial risk prediction by modeling the dialogues in earnings conference calls. In *Findings of the Association for Computational Linguistics: EMNLP 2022*, pages 1623–1633, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.

Ramit Sawhney, Piyush Khanna, Arshiya Aggarwal, Taru Jain, Puneet Mathur, and Rajiv Ratn Shah. 2020a. VolTAGE: Volatility forecasting via text audio fusion with graph convolution networks for earnings calls. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 8001–8013. Association for Computational Linguistics.

Ramit Sawhney, Puneet Mathur, Ayush Mangal, Piyush Khanna, Rajiv Ratn Shah, and Roger Zimmermann. 2020b. Multimodal multi-task financial risk forecasting. In *Proceedings of the 28th ACM International Conference on Multimedia (MM)*, pages 456–465.

Ming-Xuan Shi, Chung-Chi Chen, Hen-Hsen Huang, and Hsin-Hsi Chen. 2023. Enhancing volatility forecasting in financial markets: A general numeral attachment dataset for understanding earnings calls. In *Proceedings of the 13th International Joint Conference on Natural Language Processing and the 3rd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 37–42, Nusa Dua, Bali. Association for Computational Linguistics.

Dong Shu, Yanguang Liu, Huopu Zhang, and Mengnan Du. 2026a. FinCall-Surprise: A large scale multimodal benchmark for earning surprise prediction. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1:*

Long Papers), pages 13357–13370. Association for Computational Linguistics.

Dong Shu, Yanguang Liu, Huopu Zhang, Saisai Hu, Haiyan Zhao, Hekun Huang, and Mengnan Du. 2026b. MM-FinEval: A multi-task multi-modal benchmark for real-world financial forecasting. arXiv:2609.38523.

Nikita Tatarinov, Siddhant Sukhani, Agam Shah, and Sudheer Chava. 2026. Language modeling for the future of finance: A survey into metrics, tasks, and data opportunities. In *Proceedings of the Fifth Workshop on Generation, Evaluation and Metrics (GEM)*, pages 718–744. Association for Computational Linguistics. ArXiv:2504.07274.

Shengkun Wang, Taoran Ji, Jianfeng He, Mariam AL-Mutairi, Dan Wang, Linhan Wang, Min Zhang, and Chang-Tien Lu. 2024. AMA-LSTM: Pioneering robust and fair financial audio analysis for stock volatility prediction. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 6: Industry Track)*, pages 379–386, Mexico City, Mexico. Association for Computational Linguistics.

Yutong Yan, Raphael Tang, Zhenyu Gao, Wenxi Jiang, and Yao Lu. 2026. DatedGPT: Preventing lookahead bias in large language models with time-aware pretraining. arXiv:2603.11838.

Linyi Yang, Jiazheng Li, Ruihai Dong, Yue Zhang, and Barry Smyth. 2022. NumHTML: Numeric-oriented hierarchical transformer model for multi-task financial forecasting. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 36, pages 11604–11612.

Linyi Yang, Yingpeng Ma, and Yue Zhang. 2023. Measuring consistency in text-based financial forecasting models. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13751–13765, Toronto, Canada. Association for Computational Linguistics.

Linyi Yang, Tin Lok James Ng, Barry Smyth, and Ruihai Dong. 2020. HTML: Hierarchical transformer-based multi-task learning for volatility prediction. In *Proceedings of The Web Conference 2020 (WWW)*, pages 441–451.

Ding Yu, Zhuo Liu, and Hangfeng He. 2025. Same company, same signal: The role of identity in earnings call transcripts. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 18403–18422, Vienna, Austria. Association for Computational Linguistics.

Ding Yu, Zhuo Liu, Hao Zhang, and Hangfeng He. 2026. Fast numbers, slow language: Bridging quantitative and qualitative earnings signals. arXiv:2606.29734.

A Extended Results

This appendix holds the full set of models behind each main-text table, the horizon-definition comparison, the reproduction study in detail, the Earnings25 bitrate study, the call-date-anchor variant of the post-cutoff test, and the code-availability audit. The complete generated grids (every corpus, target, metric, split and horizon) are in the separate supplement. Every number traces to a released result CSV with a run record, and the curated tables copy cells from those CSVs (`REPRODUCE.md` lists the command behind each). Asterisks in Tables T1, T2 and T3 mark Diebold–Mariano significance, against persistence in the first and against the Stage-1 GBDT in the other two. In Tables 2 and T4 they mark the margin test.

A.1 All models of Stages 0–3

Figure S1 shows the modeling stages of the evaluation pipeline.

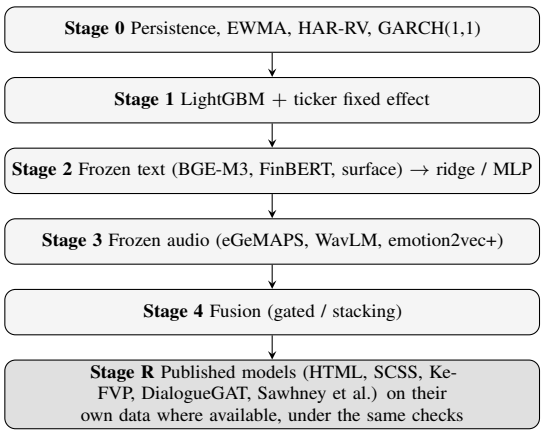


Figure S1: **The modeling ladder.** From Stage 2 up, every model also carries past-volatility covariates (and is reported with them ablated). Every stage runs through the identity checks. The darker box is the reproduction track, which applies the same checks to published models on their own data.

Tables T1, T2 and T3 give the full Stage-0/1, Stage-2 and Stage-3 model sets on the Δv target, including the past-volatility-only and MLP heads omitted from Table 2.

Figure S2 plots the canonical model of each stage from Table 2 against the train-mean constant and the past-volatility ridge S0′.

A.2 Increment over the price-only reference

Table T4 gives the paired comparison, fixed in advance, behind the markers of Table 2, on all three splits and under both the measured target and the forward-looking one. The forward-looking target removes most of the positive drift of Δv (its mean at τ=3 falls from 0.51 to 0.12). Its only improvement over the reference is the stack at τ=30 on the ticker-disjoint split, which is not in date order. No content model improves on the date-ordered company-disjoint split.

Model	τ=3	τ=7	τ=15	τ=30
Temporal split (n=483)				
Persistence	0.000	0.000	0.000	0.000
EWMA	0.465*	0.389*	0.259*	0.019
HAR-RV	0.444*	0.372*	0.138*	-0.285*
GARCH(1,1)	0.472*	0.463*	0.298*	0.023
GBDT (ticker FE)	0.340*	0.249*	-0.126	-0.948*
Ticker-disjoint split (n=484)				
Persistence	0.000	0.000	0.000	0.000
EWMA	0.520*	0.379*	0.142*	0.112*
HAR-RV	0.570*	0.491*	0.349*	0.345*
GARCH(1,1)	0.561*	0.484*	0.334*	0.298*
GBDT (ticker FE)	0.494*	0.440*	0.292*	0.237*

Table T1: **Baselines.** R^2_{OOS} vs. persistence on the Δv target, FinCall-Surprise (test). HAR-RV is strong except on the COVID-regime temporal τ=30 cell. Asterisks mark Diebold–Mariano significance vs. persistence ($p < 0.05$).

Model	τ=3	τ=7	τ=15	τ=30
Temporal split (n=483)				
Ridge (text)	0.182*	0.202	0.044*	-0.133*
Ridge (past-vol)	0.412*	0.314*	0.094*	0.113*
Ridge (text+vol)	0.333	0.285	0.080*	-0.077*
MLP (text)	0.013*	0.069*	-0.253	-1.202
Ticker-disjoint split (n=484)				
Ridge (text)	0.195*	0.217*	0.195*	0.169
Ridge (past-vol)	0.552*	0.469	0.345	0.318*
Ridge (text+vol)	0.399*	0.375	0.256	0.301
MLP (past-vol)	0.577*	0.508*	0.381*	0.356*

Table T2: **Text content (Stage 2).** R^2_{OOS} vs. persistence on the Δv target, FinCall-Surprise (test). The text-only head does not separate from the past-volatility head, and adding text to past volatility does not help. Asterisks mark uncorrected Diebold–Mariano significance vs. the Stage-1 ticker-FE GBDT ($p < 0.05$). Full head set and the Holm-corrected tests in the supplement.

A.3 The released splits and the toolkit

Table T5 gives the size, company count, company overlap and date range of every split. Table T6 states what each check in the toolkit needs and what it can show.

A.4 Implementation details

The identity probe is a multinomial logistic regression (C=1, standardized inputs) fit on a ran-

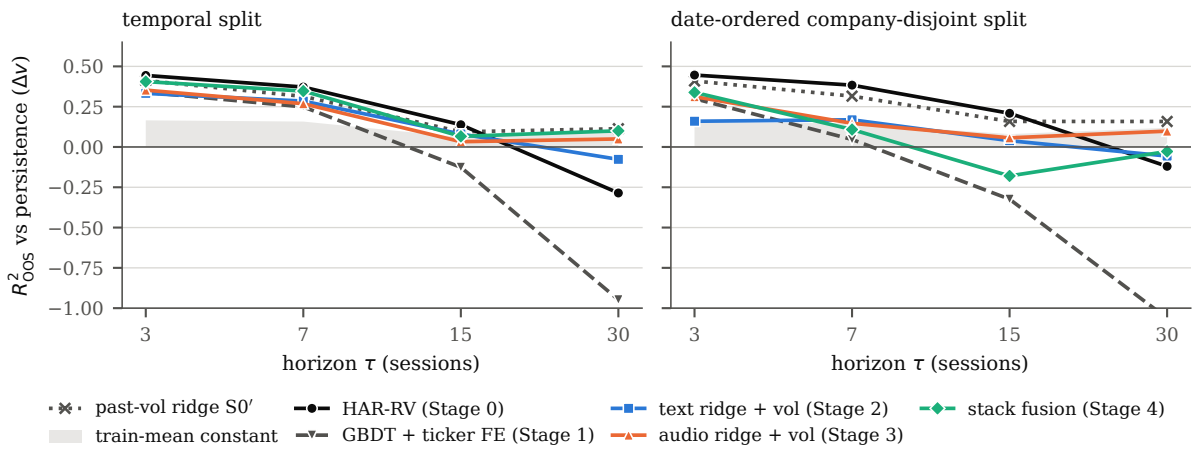


Figure S2: **The ladder.** R^2_{OOS} vs. persistence on Δv (FinCall test) for the canonical model of each stage, with the past-volatility ridge S0′ as a dotted line. The shaded band is the train-mean constant. Content stages clear HAR-RV only at τ=30, where HAR-RV itself falls below persistence on both splits, and none clears S0′ there. Data from Table 2.

dom 70/30 split (seed 0) of the calls whose company has at least two calls. The swaps permute the test calls' text or audio features among the test calls, within each company for the same-company swap and across all test calls for the random swap, over five seeds. The encoders are BAAI/bge-m3, ProsusAI/finbert, microsoft/wavlm-large and iic/emotion2vec_plus_large as published on their hubs at extraction time. The released feature archives carry content hashes, and the post-cutoff test scores exactly the features and heads those archives contain, so it does not depend on re-downloading a checkpoint.

A.5 Horizon definition

Table T7 compares the trading-day and calendar-day definitions of the horizon on FinCall and MAEC, both at the level of the targets (sessions per window, correlation between the two definitions) and at the level of the Stage-0 results (persistence MSE and HAR-RV R^2_{OOS} under each definition).

A.6 Reproduction study

Three kinds of run appear here. Direct reproductions use shipped data, repaired runs use patched code or rebuilt inputs, and ports move a model onto another corpus where the original was never released. Table T8 is the first entry of the study, the HTML head over our frozen BGE-M3 chunk sequences on EC under the published split, a gapped split and a ticker-disjoint split with identical labels. Table T9 reruns the five papers with usable code on their own data with the authors' code wherever it runs and a line-by-line port where it cannot. The released fidelity ledger records every patch and substitution. Table T10 gives the absolute values behind Table 4, and Table T13 the code-availability audit of the seventeen papers.

Same-Company-Same-Signal reproduces. Its TSMixer baseline matches the 76 published cells to a median 10^{-5} from the shipped data alone. Its TMLP text model on the authors' OpenAI embeddings matches per-window means within 0.03 and recovers the paper's finding that embeddings swapped within a company do as well as the real ones. Its training-free baselines on EC and MAEC reproduce all 48 published cells at the notebook's rounding. **HTML** does not. The released driver splits features and labels with three independent unseeded shuffles. With the authors' model classes unchanged on rebuilt inputs, the published recipe (ten epochs, learning rate 2×10^{-5}, raw log-volatility targets) beats the published number only at τ=3 and τ=7 on the shipped code's random split with dropout 0, reaches the published 15- and 30-day values on no split, and on the VolTAGE split every later paper inherited it is worse than persistence at every horizon except τ=3. Its text+audio row reproduces under the shipped code's random split and under no other. **Sawhney et al.** cannot run as released (an unshipped call inventory, price files and lexicons, and a TensorFlow 2.1 stack). Ported, the finance SVR alone gives 0.712/0.370/0.218/0.154, the text and audio branches are untrained after the two and one epochs the scripts prescribe, and the published ensemble is reproduced only when the

Model	τ=3	τ=7	τ=15	τ=30
Temporal split (n=483)				
Ridge (eGeMAPS)	0.163*	0.156*	0.059*	0.079*
Ridge (WavLM)	0.156*	0.148*	-0.027	-0.008*
Ridge (emotion2vec+)	0.155*	0.129*	0.022*	0.027*
Ridge (WavLM+past-vol)	0.357	0.262	-0.044	0.052*
Ticker-disjoint split (n=484)				
Ridge (eGeMAPS)	0.203*	0.119*	0.137*	0.044*
Ridge (WavLM)	0.178*	0.192*	0.131*	0.026*
Ridge (emotion2vec+)	0.181*	0.185*	0.122*	0.047*
Ridge (WavLM+past-vol)	0.456	0.390	0.267	0.222
Audio-shuffle control of WavLM+past-vol (ticker-disjoint)				
Real audio	0.456	0.390	0.267	0.222
Within-ticker shuffle	0.453	0.373	0.229	0.152
Global shuffle	0.453	0.323	0.164	0.099

Table T3: **Audio content (Stage 3).** R^2_{OOS} vs. persistence on the Δv target, FinCall-Surprise (test). Audio-only heads stay within 0.10 of the train-mean constant and never beat it by more than 0.02. The head with past volatility is below the past-volatility-only ridge of Table T2 in every cell, and at τ=3 it keeps its score when every test call receives a random other company's audio (bottom block), so past volatility rather than audio content carries it. Asterisks mark uncorrected Diebold–Mariano significance vs. the Stage-1 ticker-FE GBDT ($p < 0.05$, read the direction from the value). Full head set and the Holm-corrected tests in the supplement.

mixing weights are chosen on the test set, as the released script does, where an untrained branch's near-constant output acts as an intercept on the finance branch. **DialogueGAT**'s corpus and labels were never released. Ported onto FinCall's speaker turns under the paper's per-year protocol, it beats persistence at short horizons and collapses on the 2020 split. **KeFVP**'s released script needed five release-defect patches before it would start. Its MAEC rows use sentence embeddings regenerated with the authors' own generator because the released ones were never published, and seven of eight cells fall within one published standard deviation (the eighth, MAEC-16 at τ=15, is the row of their table that does not average to its own stated mean). On EC the released KePt embeddings are unavailable upstream (the linked folder is empty). With BERT-large pooler embeddings from the same generator, which is KeFVP without its knowledge-enhanced pre-training, the mean test MSE is 0.651/0.302/0.194/0.127 against the published 0.610/0.291/0.183/0.114, above all four published means and within one of our repeat standard deviations at only two horizons, so the row is not counted as reproduced.

Under the checks, the 30-session gap removes half to two thirds of KeFVP's MAEC training calls, so those rows cannot separate leakage from data loss and are left to the ledger. DialogueGAT's port (one seed, 42–60 held-out calls per year) shows no consistent direction under either check.

A.7 Audio by bitrate on Earnings25

Table T11 reruns the Stage-3 audio models on Earnings25, pooled and per bitrate stratum, with the random swap as the check.

A.8 Post-cutoff evaluation under the call-date anchor

Table T12 repeats the post-cutoff evaluation with day 0 anchored on the call date on both sides of the cutoff, the anchor every released label set uses, with the heads refit under that anchor. HAR-RV barely moves, but the ridge models that carry a learned past-volatility block collapse before the cutoff under that anchor, because the block absorbs the reaction session that the after-hours rule leaves inside the pre-window for before-open calls. On Δv at τ=30 the frozen WavLM+past-volatility model is the only content model whose post-cutoff confidence interval excludes zero under the measured anchor (0.18, CI 0.14–0.24, against HAR's 0.08, and the constant also scores 0.18 there). That is a single-quarter observation without a Diebold–Mariano test and we report it as exploratory.

A.9 Code availability

Table T13 is the code-availability audit of the seventeen papers in this line, regenerated by `ecvol audit code`. The papers are those that predict a financial outcome from earnings calls and were cited in the volatility line, so they include two direction-prediction models and one consistency test.

A.10 The reporting checklist

Table T14 expands the six items of §7 with the evidence behind each and the part of the release that provides it.

Target, split	Model	τ=3	τ=7	τ=15	τ=30
Measured anchor, temporal	HAR-RV	+5 [+3, +8]*	+8 [+6, +11]*	+5 [+2, +8]*	−45 [−61, −31]†
	S2 text ridge + vol	−14 [−25, −3]†	−4 [−15, +6]	−2 [−12, +8]	−21 [−32, −12]†
	S3 audio ridge + vol	−10 [−18, −3]†	−7 [−15, +0]†	−7 [−14, −1]†	−7 [−12, −2]†
	S4 stack fusion	−1 [−7, +4]	+5 [−0, +10]	−3 [−9, +2]	−1 [−6, +4]
Measured anchor, date-ordered disjoint	HAR-RV	+6 [+0, +11]	+10 [+3, +16]*	+6 [−6, +17]	−33 [−64, −7]†
	S2 text ridge + vol	−42 [−90, −12]†	−21 [−45, −4]†	−14 [−34, +1]†	−25 [−49, −5]†
	S3 audio ridge + vol	−16 [−38, −2]†	−25 [−42, −10]†	−12 [−24, −1]†	−7 [−21, +6]
	S4 stack fusion	−12 [−32, +0]†	−30 [−50, −16]†	−40 [−64, −21]†	−22 [−45, −3]†
Measured anchor, ticker-disjoint	HAR-RV	+4 [+0, +8]	+4 [+1, +8]	+1 [−3, +4]	+4 [−0, +7]
	S2 text ridge + vol	−34 [−53, −20]†	−18 [−29, −6]†	−14 [−26, −4]†	−2 [−16, +8]
	S3 audio ridge + vol	−25 [−45, −9]†	−20 [−37, −6]†	−16 [−33, −3]†	−15 [−31, +1]†
	S4 stack fusion	−5 [−13, +2]	−1 [−11, +8]	−1 [−13, +9]	+11 [+1, +20]
Forward-looking, temporal	HAR-RV	+12 [+9, +14]*	+14 [+11, +17]*	+7 [+3, +10]*	−67 [−87, −49]†
	S2 text ridge + vol	−32 [−47, −19]†	−23 [−35, −11]†	−10 [−21, −0]†	−35 [−48, −23]†
	S3 audio ridge + vol	−24 [−47, −8]†	−20 [−32, −10]†	−10 [−19, −2]†	−5 [−11, +1]†
	S4 stack fusion	−3 [−9, +3]	−5 [−12, +1]†	−18 [−27, −10]†	−1 [−6, +4]
Forward-looking, date-ordered disjoint	HAR-RV	+9 [+2, +16]	+12 [+5, +19]*	+14 [+4, +22]*	−34 [−68, −6]†
	S2 text ridge + vol	−63 [−112, −30]†	−34 [−60, −12]†	−24 [−39, −12]†	−32 [−54, −12]†
	S3 audio ridge + vol	−32 [−63, −12]†	−33 [−58, −13]†	−16 [−28, −5]†	−5 [−20, +7]
	S4 stack fusion	−13 [−35, +4]	−75 [−115, −43]†	−97 [−134, −64]†	−25 [−46, −7]†
Forward-looking, ticker-disjoint	HAR-RV	+4 [−1, +8]	+3 [−2, +8]	−1 [−6, +3]	+4 [−0, +7]
	S2 text ridge + vol	−47 [−65, −31]†	−18 [−32, −7]†	+0 [−13, +11]	+7 [−3, +16]
	S3 audio ridge + vol	−25 [−39, −13]†	−16 [−31, −5]†	−7 [−15, −0]†	−13 [−28, +0]†
	S4 stack fusion	−4 [−13, +3]	−2 [−12, +6]	+9 [+1, +15]	+16 [+8, +23]*

Table T4: **Increment over the price-only reference.** Relative reduction of test MSE on Δv (%) against the past-volatility ridge on the same calls, with a 95% percentile bootstrap interval that resamples companies (1000 draws). Positive values favor the content model. The 2% margin and the reading rule were fixed before the table was computed. A dagger marks an upper bound below the margin (no meaningful benefit), an asterisk a lower bound above it (improvement), and unmarked cells are inconclusive. HAR-RV is shown against the same reference. The forward-looking target starts the post window one session later, so no target return begins before the call has ended.

Corpus	Split	Train	Val	Test	Test companies	Shared train/test	Train dates	Test dates
FinCall	temporal	1489	228	483	315	287	2019-01 to 2021-01	2021-07 to 2021-12
FinCall	ticker-disjoint	1706	243	484	78	0	2019-01 to 2021-12	2019-01 to 2021-12
FinCall	date-ordered disjoint	1042	22	97	64	0	2019-01 to 2021-01	2021-07 to 2021-12
MAEC	temporal	1666	69	529	358	275	2015-02 to 2016-11	2017-05 to 2018-06
MAEC	ticker-disjoint	1884	271	533	198	0	2015-02 to 2018-06	2015-02 to 2018-05
MAEC	date-ordered disjoint	1159	5	104	71	0	2015-02 to 2016-11	2017-05 to 2018-05

Table T5: **The released splits.** Calls per segment, distinct test companies, companies shared between training and test, and the date ranges of the training and test segments (a few MAEC calls lack a 30-session target and drop from the τ=30 cells). The temporal splits drop the calls in a 30-session gap before validation and before test, so every training label has matured before the test origin. In the company-disjoint splits validation and test companies are also disjoint from each other and from training. The date-ordered company-disjoint split keeps the temporal boundaries and removes every test company from training, which leaves few calls, including only 22 FinCall and 5 MAEC validation calls. Nine FinCall training calls carry a pre-2019 date that their transcripts contradict and are excluded from the date ranges.

Check	Needs	Shows
Prediction diagnostics	predictions, outcomes, ticker, date, split	scores by seen and unseen company, company share of prediction variance, split integrity
Identity probe	call embeddings, tickers	whether identity is decodable
Content swaps	the model and its inputs	how predictions respond to another call with price inputs fixed
Company-disjoint retraining	the training pipeline and splits	what is learned without company history

Table T6: **The toolkit.** Only the first row runs on a prediction file. A permutation of existing predictions moves their price inputs along with the content, so it is a descriptive diagnostic, not a content intervention.

	τ=3	τ=7	τ=15	τ=30
Targets (FinCall, calls ok under both conventions)				
Sessions in calendar window (mean)	2.4	4.9	10.8	21.2
Calls with $\geq$2 sessions (of 2,433)	1,398	2,433	2,433	2,433
corr($v_{\text{post}}^{\text{trad}}, v_{\text{post}}^{\text{cal}}$)	0.784	0.942	0.963	0.940
corr($\Delta v^{\text{trad}}, \Delta v^{\text{cal}}$)	0.751	0.887	0.904	0.891
Persistence MSE, level-v, temporal test (trading / calendar)				
FinCall	1.270 / 2.071	0.448 / 0.612	0.211 / 0.298	0.115 / 0.152
MAEC	1.774 / 2.553	0.712 / 0.986	0.356 / 0.492	0.201 / 0.263
HAR-RV R^2_{OOS} vs. persistence, level-v, temporal test (trading / calendar)				
FinCall	0.444 / 0.419	0.372 / 0.402	0.138 / 0.275	-0.285 / -0.053
MAEC	0.603 / 0.562	0.486 / 0.502	0.321 / 0.400	0.221 / 0.265

Table T7: Horizon definition. The calendar-day windows of Qin and Yang (2019) hold about 0.7τ sessions, and at τ=3 43% of the calls have at most one post-call session and are excluded. The persistence MSE, the leaderboard's metric, rises by 44–63% at τ=3 under calendar days, so published MSE values are not comparable across definitions. The FinCall temporal τ=30 deficit of HAR-RV against persistence (the COVID-regime cell, §5) holds under both definitions (-0.285 and -0.053). The full model and horizon grid under both definitions is in the supplement.

Condition (EC)	model	τ=3	τ=7	τ=15	τ=30
paper, published split, published labels	HTML (text, WWM-BERT)	1.175	0.372	0.153	0.133
published split, published labels	HTML head (BGE-M3)	0.769	0.436	0.285	0.223
	persistence	1.485	0.570	0.340	0.202
published split, our labels	HTML head (BGE-M3)	0.760	0.420	0.285	0.214
	persistence	1.826	0.590	0.333	0.183
embargoed temporal, our labels	HTML head (BGE-M3)	0.778	0.448	0.278	0.265
	persistence	1.948	0.629	0.353	0.192
ticker-disjoint, our labels	HTML head (BGE-M3)	**0.773**	**0.503**	**0.394**	**0.291**
	persistence	1.670	0.728	0.418	0.213

Table T8: **Reproduced prior model under temporal and company-disjoint controls.** The HTML head reproduces the paper's ordering against price baselines at τ=3 and beats the published number there, but not the published τ=15 and τ=30 values, which lie below the persistence baseline of the shipped labels themselves. At τ=30 every condition is worse than persistence. Moving the same head and labels from the published split to a ticker-disjoint split raises test MSE by 2%, 20%, 38% and 36%. The embargoed temporal split costs 2%, 7%, -2% and 24%.

Paper / setting	what ran	ours				published			
		τ=3	7	15	30	τ=3	7	15	30
HTML (Yang et al., 2020), EC, authors' model classes verbatim, inputs rebuilt (WWM-BERT layer −2), driver corrected									
as coded, random 70/10/20, dropout 0	text	0.660	0.336	0.253	0.160	1.175	0.372	0.153	0.133
as published, chronological 70/10/20, dropout 0.5	text	0.972	0.439	0.406	0.248				
published split (VolTAGE), dropout 0.5	text	1.231	1.122	1.063	1.479				
persistence on the shipped labels	n/a	1.485	0.570	0.340	0.202				
as coded, random 70/10/20, dropout 0	text + 27 Praat	0.658	0.347	0.244	0.165	0.845	0.349	0.251	0.158
as published, chronological 70/10/20, dropout 0.5	text + 27 Praat	0.876	0.520	0.291	0.239				
published split (VolTAGE), dropout 0.5	text + 27 Praat	2.701	1.195	1.436	1.411				
Same-Company-Same-Signal (Yu et al., 2025), DEC, authors' `run.py`*, per-window means over 19 rolling quarters*									
TSMixer (price history only)	76 cells	0.561	0.252	0.210	0.257	0.561	0.252	0.211	0.256
TMLP on OpenAI embeddings	76 cells	0.608	0.285	0.215	0.202	0.587	0.283	0.211	0.205
TMLP, embeddings shuffled within ticker	76 cells	0.572	0.276	0.213	0.209	0.565	0.273	0.204	0.200
TMLP, embeddings shuffled across all	76 cells	0.646	0.315	0.259	0.249	0.637	0.319	0.255	0.247
Same-Company-Same-Signal, EC / MAEC, authors' notebook, training-free, augmented same-ticker earnings history									
EC, Aug_STPEV	no training	0.569	0.293	0.201	0.122	0.569	0.293	0.201	0.122
MAEC-15, Aug_STPEV	no training	0.443	0.214	0.130	0.112	0.443	0.214	0.130	0.112
MAEC-16, Aug_STPEV	no training	0.470	0.238	0.168	0.113	0.470	0.238	0.168	0.113
KeFVP (Niu et al., 2023), authors' `final_series_infer.py` *(five release-defect patches), 10 repeats*									
MAEC-15, regenerated BERT-base embeddings	text + price	0.419	0.185	0.123	0.086	0.418	0.187	0.122	0.087
MAEC-16, regenerated BERT-base embeddings	text + price	0.442	0.288	0.363	0.188	0.445	0.279	0.303	0.177
EC, regenerated BERT-large embeddings (no KePt)	text + price	0.651	0.302	0.194	0.127	0.610	0.291	0.183	0.114
Sawhney et al. (2020b), EC, PyTorch port of the released scripts (their labels, split, branches); 5 seeds									
finance SVR (30 past volatilities)	port	0.712	0.370	0.218	0.154				
text BiLSTM, their 2 epochs	port	2.135	1.580	1.384	1.401				
aligned audio, their 1 epoch	port	6.514	5.295	5.368	5.130				
ensemble, (α, β) tuned on test (theirs)	port	0.639	0.298	0.175	0.102	0.601	0.308	0.181	0.119
ensemble, (α, β) tuned on validation	port	0.686	0.324	0.187	0.117				
persistence on their labels	n/a	1.491	0.563	0.345	0.205				
DialogueGAT (Sang and Bao, 2022), PyG port on FinCall turns (role-typed speakers); published cells are on their unreleased corpus									
FinCall 2019 (their 2015 protocol)	port	0.694	0.320	0.206	0.134	0.453	0.324	0.190	n/a
persistence	n/a	1.429	0.566	0.289	0.139				
FinCall 2020 (their 2016 protocol)	port	0.621	0.350	0.392	0.270	0.455	0.288	0.181	n/a
persistence	n/a	1.409	0.335	0.184	0.101				
FinCall 2021 (their 2017–18 protocol)	port	0.794	0.344	0.182	0.082	0.409	0.289	0.204	n/a
persistence	n/a	1.301	0.465	0.225	0.118				

Table T9: **Reproductions, direct, repaired and ported.** Every cell regenerates from one `ecvol reproduce` command, and the fidelity ledger lists what is verbatim, patched, substituted or reason-coded per paper. TSMixer matches to 10^{-5}, TMLP within single-seed noise (including the paper's finding that real and within-company shuffled embeddings score alike), and KeFVP on MAEC within one published standard deviation on seven of eight cells with regenerated sentence embeddings. On EC, whose released KePt embeddings are unavailable upstream, the regenerated BERT-large row is above all four published means and within one of our repeat standard deviations at only two horizons, and is not counted. HTML's text-only recipe does not reach its published long-horizon cells on any split, while its text+audio row reproduces at $\tau \geq 7$ under the shipped-code condition (random split, dropout 0) and on no other. Sawhney's published ensemble is reachable only when the mixing weights are tuned on the test set, as the released code does, with untrained text and audio branches acting as an intercept on the finance SVR.

Model / condition	n_{tr}	n_{te}	τ=3	7	15	30
HTML, authors' classes (EC; seeds 0–2)						
text, shipped-code random split (anchor)	≈390	112	0.660	0.336	0.253	0.160
text, embargoed	285	101	0.726	0.347	0.246	0.178
text, ticker-disjoint	386	109	0.777	0.522	0.393	0.291
persistence on the ticker-disjoint test set	n/a	109	1.452	0.550	0.327	0.188
text+audio, anchor	≈390	112	0.658	0.347	0.244	0.165
text+audio, embargoed	285	101	0.684	0.345	0.232	0.186
text+audio, ticker-disjoint	386	109	0.789	0.500	0.394	0.281
transcript shuffle, anchor split, the 97 test calls whose company has another call						
text, real transcript	≈390	97	0.683	0.348	0.262	0.165
text, another call of the *same* company	≈390	97	0.693	0.361	0.269	0.169
text, a call of a *different* company	≈390	97	0.745	0.386	0.318	0.203
text+audio, real / same company / different company, τ=3	≈390	97	0.675 / 0.692 / 0.784			
Sawhney et al., port, ensemble with (α, β) tuned on validation (EC; 5 seeds)						
their split (anchor)	328	110	0.686	0.324	0.187	0.117
embargoed	283	100	0.719	0.306	0.184	0.107
ticker-disjoint	383	109	0.775	0.416	0.263	0.161
its own finance SVR, ticker-disjoint	383	109	0.700	0.365	0.256	0.158
Same-Company-Same-Signal, authors' `run.py` *(DEC; seed 2021; means over 19 rolling quarters)*						
TMLP (text), paired anchor	900	30	0.591	0.288	0.224	0.211
TMLP (text), ticker-disjoint	600	30	0.660	0.354	0.280	0.267
TMLP (text), embargoed (vs anchor 0.608 / 0.285 / 0.215 / 0.202)	895	90	0.585	0.287	0.216	0.208
TSMixer (price), paired anchor	900	30	0.519	0.241	0.206	0.252
TSMixer (price), ticker-disjoint	600	30	0.556	0.254	0.194	0.233
persistence on the held-out calls	n/a	30	1.258	0.457	0.252	0.224
KeFVP, authors' script (MAEC, regenerated BERT-base embeddings; 3 repeats)						
MAEC-15, paired anchor	535	48	0.470	0.192	0.145	0.114
MAEC-15, ticker-disjoint	353	48	0.458	0.199	0.150	0.113
MAEC-16, paired anchor	980	92	0.416	0.248	0.347	0.186
MAEC-16, ticker-disjoint	661	92	0.378	0.241	0.312	0.178
persistence on the MAEC-16 held-out calls	n/a	92	0.622	0.265	0.170	0.116

Table T10: **The reproduced models under our controls.** Each block keeps the configuration of Table T9 and changes only the split. Paired anchors score the unchanged model on exactly the ticker-disjoint test calls. Removing same-company history costs the three text models 12–82% (HTML, Sawhney, TMLP, where TMLP is worse in 16–18 of 19 quarters), while both price-dominated models (TSMixer, KeFVP) show no consistent change, although the ticker-disjoint training sets are as large (EC) or cut by the same third (DEC, MAEC) in both groups. The 30-session embargo moves HTML, Sawhney and SCSS by at most 13% in either direction. KeFVP's embargoed MAEC re-splits keep only a half or a third of its training calls, cannot separate leakage from data loss, and are reported in the ledger rather than here. DialogueGAT (one seed, 42–60 held-out calls per year) shows no consistent direction and is also left to the ledger.

Stratum	calls (test)		τ=3	τ=7	τ=15	τ=30
All	514 (95)	real	0.435	0.328	0.171	0.220
		global shuffle	0.454	0.327	0.141	0.206
		increment	**−0.019**	**0.001**	**0.030**	**0.014**
64 kbps	281 (39)	real	0.370	0.250	0.109	0.310
		global shuffle	0.383	0.266	0.133	0.303
		increment	**−0.013**	**−0.016**	**−0.024**	**0.007**
≤24 kbps	232 (55)	real	0.343	0.328	0.174	0.145
		global shuffle	0.443	0.354	0.174	0.128
		increment	**−0.100**	**−0.026**	**−0.001**	**0.018**

Table T11: **Audio on Earnings25 by bitrate stratum.** R^2_{OOS} vs. persistence on Δv for the ridge on WavLM plus past volatility, ticker-disjoint test. With one call per company the same-company swap is the identity, so the random swap is the check. Call counts are inventory calls with audio at that bitrate (one call at 32 kbps sits in neither stratum). The heads see the 475 admitted, priced calls, and the test count is their ticker-disjoint segment. The audio increment never exceeds 0.03, is negative in 7 of the 12 cells, and has no consistent sign across strata. The 64 kbps stratum is not better than the ≤24 kbps one. The full grid (eGeMAPS, emotion2vec+, fusion, MLP heads, all targets) is in `result_table_3_earnings25.csv`.

Head	seg.	τ=3	τ=7	τ=15	τ=30
HAR-RV	in	$0.44 \to 0.46$	$0.37 \to 0.41$	$0.14 \to 0.21$	$-0.29 \to -0.28$
	post	$0.60 \to 0.56$	$0.54 \to 0.51$	$0.35 \to 0.33$	$0.08 \to -0.12$
text + past-vol	in	$0.33 \to -1.71$	$0.28 \to -3.12$	$0.08 \to -3.62$	$-0.08 \to -0.47$
	post	$0.31 \to 0.28$	$0.33 \to 0.27$	$0.23 \to 0.15$	$0.12 \to -0.09$
WavLM + past-vol	in	$0.36 \to 0.26$	$0.26 \to -1.02$	$-0.04 \to -2.01$	$0.05 \to -0.56$
	post	$0.50 \to 0.43$	$0.43 \to 0.32$	$0.15 \to 0.06$	$0.18 \to 0.05$
fusion + past-vol	in	$0.29 \to -1.73$	$0.22 \to -3.13$	$0.04 \to -0.34$	$-0.11 \to -0.45$
	post	$-0.83 \to -9.97$	$0.32 \to -2.29$	$0.19 \to 0.02$	$0.01 \to -0.28$

Table T12: **Anchor sensitivity.** The same models, refit and scored with day 0 on the call date, the anchor of the released label sets (Δv, R^2_{OOS} vs. persistence). Each cell reads measured $\to$ call-date. The ordering on level-v is in the CSV.

Paper	venue	probe (date)	verdict	reason
What You Say and How You Say It Matters (Qin and Yang, 2019)	ACL 2019	200, 0 code files, pushed 2022-08-04 (2026-10-12)	partial	dataset released (Drive, 5-part zip); no model code in the repository
HTML (Yang et al., 2020)	WWW 2020	200, 15 code files, pushed 2024-01-05 (2026-10-12)	partial	model classes run verbatim; the released driver imports TensorFlow-1 set_random_seed and torchtext and splits features and labels with independent unseeded shuffles; inputs (WWM-BERT .npy, Praat features) behind dead Drive links, rebuilt; run with a corrected driver
Multimodal financial forecasting (Sawhney et al., 2020b)	ACM MM 2020	200, 112 code files, pushed 2020-10-26 (2026-10-12)	partial	TF 2.1 / Keras 2.3.1 / tensorflow-addons 0.8.3 pins; unshipped call inventory, price files and two lexicons; text-feature script returns after one record; no multi-task loss in the released code; ported line by line to PyTorch
VolTAGE (Sawhney et al., 2020a)	EMNLP 2020	200, 5 code files, pushed 2021-11-16 (2026-10-12)	partial	ships the EC split/label files (byte-identical to KeFVP's) and GCN code; embeddings as .pkl; not ported
KeFVP (Niu et al., 2023)	Findings of EMNLP 2023	200, 42 code files, pushed 2024-01-28 (2026-10-12)	partial	requirements.txt does not install as written; the script starts only after five release-defect patches (missing set_seed, package init importing undefined names, a dropped class header, an unreleased 'latent' package, plotting imports); MAEC embeddings never released (regenerated with the authors' generator); EC KePt pickle Drive-only
DialogueGAT (Sang and Bao, 2022)	Findings of EMNLP 2022	200, 6 code files, pushed 2023-06-12 (2026-10-12)	partial	MIT code, tau<=15 only; training pickle, SeekingAlpha corpus and CRSP labels never released; DGL has no wheels for torch 2.11; ported (PyG) onto FinCall turns
Same Company, Same Signal (Yu et al., 2025)	Findings of ACL 2025	200, 38 code files, pushed 2026-08-13 (2026-10-12)	runnable	MIT; ships DEC with features and rolling masks; PEV/STPEV to 3 decimals, TSMixer 76 cells to 1e-5, TMLP 228 cells within single-seed noise on the authors' embeddings
FinTrust (Yang et al., 2023)	ACL 2023	200, 4 code files, pushed 2024-01-22 (2026-10-12)	partial	perturbation harness + 3/7/15/30 data; complementary control axis, not a volatility model
Sound of Risk (Chen et al., 2025)	arXiv 2025	200, 0 code files, pushed 2025-08-26 (2026-10-12)	none	repository holds figures only, zero source files; proprietary 1,795-call corpus
DeFVP (Niu et al., 2024)	ICME 2024	409 (2026-10-12)	none	repository exists but is empty (size 0 KB, never pushed; API returns 409)
ECHO-GL (Liu et al., 2024)	AAAI 2024	200, 3 code files, pushed 2024-02-26 (2026-10-12)	none	README states the code 'can not run at present'; target is direction, not volatility
ECC Analyzer (Cao et al., 2024b)	ICAIF 2024	no_url	none	no code released; GPT-4 Turbo + OpenAI embeddings
RiskLabs (Cao et al., 2024a)	arXiv 2024	no_url	none	no code released; GPT-4/3.5
NumHTML (Yang et al., 2022)	AAAI 2022	no_url	none	paper promises code; no URL given
GNA-Vol (Shi et al., 2023)	IJCNLP-AACL 2023	no_url	none	artifact URL in the paper returned 404 when checked on 2026-08-23
AMA-LSTM (Wang et al., 2024)	NAACL 2024 Industry	200, 0 code files, pushed 2024-04-08 (2026-10-12)	none	author-linked repository holds only a README at the audit date; no runnable model code
AT-FinGPT (Liu et al., 2025)	Finance Research Letters 2025	no_url	none	paywalled; no artifact

Table T13: **Code-availability audit of seventeen earnings-call prediction papers** (regenerated by `ecvol audit code` with a live GitHub probe on the date shown). One of seventeen papers ships code that runs as released. Seven are partial artifact releases, either data without model code (Qin and Yang) or code with a broken environment, missing inputs or defects that must be repaired before it starts (HTML's driver imports TensorFlow-1 symbols and decorrelates features from labels, and KeFVP needs five release-defect patches). Nine ship nothing usable. The reason column records what the faithful-reproduction sprint (Table T9) did with each.

Report	Why	Provided by
A company-disjoint split next to the temporal one	On the time-ordered splits in use, 44–80% of test calls come from companies in training, and a probe reads the company out of a frozen call embedding 89.5% of the time (Tables 1, 3).	ticker-disjoint splits for FinCall-Surprise and MAEC (0.0% overlap). The audit command reports the share of test companies seen in training and scores seen and unseen companies separately.
A gap of at least the longest horizon between training and test	Five of six shipped boundaries have a zero-day gap while targets reach 30 days past the call, so training labels contain test-period outcomes.	Released temporal splits with a 30-session gap. The audit command reports the gap in business days and counts training windows that reach into the test period.
Past-volatility baselines (persistence, HAR-RV, and a constant on change targets)	They are hard to beat and often missing from published comparisons. On the measured anchor the constant already scores 0.06–0.17 on Δv (Table 2).	Stage 0 of the harness and the floors in the audit command, including a company-only model.
Same-company and random swaps	Where companies recur in training, a same-company swap moves the text model by less than 0.03 while a random swap hurts, which is consistent with reliance on company-level information rather than the call (Table 3).	`ecvol controls` for feature-level swaps. Prediction-side swaps in the audit command for any prediction file.
Day 0 measured from the release time	Under the call-date anchor, day 0 moves for 53–72% of calls and the reaction session sits in the pre window (§3).	Release times from 8-K filings in the timing tables. Call-date targets shipped as a labeled variant for comparison with prior work.
The horizon definition, calendar or trading days	The two definitions share only 0.78 correlation at $\tau=3$ and the persistence MSE is 44–63% higher under calendar days, so an MSE without its definition cannot be compared.	Targets under both definitions in every archive and a convention column in every result table.

Table T14: **The reporting checklist of §7.** Each item with the evidence behind it and the part of the release that provides it.